Research Statement

Motivation

Language models are quickly taking on roles where people rely on what they say: answering questions, summarizing research, and supporting decisions. Yet they still make basic reasoning mistakes, often confidently. My research asks a question about this gap: when a model reasons wrongly, can we understand why, and fix the cause rather than the symptom?

Over the past two years, working largely independently, I approached this question from the outside of the model inward. Each project improved model behavior, and each one also showed me the limits of improving behavior without understanding the mechanism behind it. That lesson is what now draws me to a PhD in interpretability and alignment.

Correcting outputs from the outside

My first project addressed hallucination through grounding. I built a retrieval-augmented question-answering system over a corpus of peer-reviewed Springer publications. Every answer is tied to retrieved papers and returned with citations and DOIs, so users can check the source themselves. Grounding limited fabricated claims, but it could not help when a model reasoned incorrectly from correct premises.

That gap led me to logical reasoning. Holliday, and my own tests on showed that even frontier models fail systematically on simple inferences with conditionals and the modals “must” and “might” [1]. For example, models conclude that Mary was not at the wedding from “If Mary was there, Sue must have been” and “It is not the case that Sue must have been there.” They treat a denial of necessity as a denial of fact.

In my master’s thesis, I developed LogiCue: metacognitive prompts that target the specific logical error behind each of the seven most difficult patterns [2]. Across models ranging from small open models to frontier systems, LogiCue substantially reduced these errors, while standard chain-of-thought prompting did not. Still, the results raised a question I could not answer from outputs alone. In one case, a model restated the prompt’s correct counterexample and then reasoned its way back to the wrong answer. Prompting changed what models said, but I could not tell whether it changed how they reasoned.

Predicting failures, and what that revealed

The natural next step was to fine-tune a model to internalize these cues, so that it would avoid the errors without being prompted. Before doing so, in work published at KI 2026, I asked whether failures could be predicted at all [3]. I trained classifiers to predict, from the structure of a problem alone, whether a model would fail on it, before the model is ever queried. Failures turned out to be highly predictable, especially on patterns involving "must," because the underlying error is so consistent.

The most important result was unexpected. After LogiCue, the failure signature the classifier had learned no longer applied; on several patterns, its predictions were even reversed. The intervention reduced errors, but the errors that survived had a different structure from the ones it removed.

This finding changed how I approached fine-tuning. A model trained to absorb these cues might fail less often, but in ways that existing checks are not designed to catch. Measuring that shift is possible from the outside; explaining it is not. Before training these cues into a model, I wanted to understand what an intervention actually changes inside it, and that question led me to interpretability.

Moving inside the model

I first tested whether logical rules could be built into a model rather than prompted. MeMo is a language model architecture that stores associations explicitly in correlation matrix memories [5]. Using a dataset I expanded from Holliday et al.'s patterns, I encoded each inference pattern as a key–value association [4]. The memory reliably retrieved the correct judgment for examples it had never seen.

The failure case was more instructive. When I stored valid inferences alongside their invalid modal versions, the memory could not tell them apart. Their representations were nearly the same, because they shared vocabulary and differed mainly in the word "must" or "might." The representations captured surface form, not logical structure. I presented this work at the MeMo Workshop on Mechanistic Interpretability and Neuro-symbolic Approaches by Design (Rome, 2026).

Most recently, I worked directly with a transformer's internal activations. I built an English-to-German steering vector for Llama-3.2-1B-Instruct, following activation addition [6]. I then tested whether it transferred across text domains, a question recent steering benchmarks leave open [7]. The vector behaved similarly across domains, but only inside a narrow window of strength: below it there was no effect, and above it outputs collapsed into incoherence. This

small experiment showed me both the power of representation-level interventions and how fragile they are without a mechanistic account of what they change.

Across these projects, one lesson repeated. Interventions on behavior succeeded partially, then failed in ways I could not explain from the outside. Explaining them requires understanding representations, which is the core question of interpretability.

Proposed research

My future research centers on one question that runs through all of my work so far: when we change a model's behavior, what actually changes inside it? Prompting, steering, and fine-tuning are widely used to make models more reliable, yet they are usually judged only by their outputs. My projects showed that an intervention can fix errors on the surface while leaving the underlying computation, and the failures that remain, poorly understood.

Understanding failures at their source. When a model reasons incorrectly, is the information it needs missing from its representations, or present but ignored by later computation? Probing and activation patching can separate these cases. Controlled tasks, where a small change in the input flips the correct answer, make this analysis causal rather than correlational.

What interventions change. Comparing a model's internal states before and after prompting, steering, or fine-tuning can show whether an intervention changes the underlying computation or only the final answer. Cases where a model states the correct reasoning and still answers wrongly connect this directly to alignment questions about whether model explanations are faithful.

Interventions whose failures stay visible. My work suggests that reducing errors can change their structure, making the remaining ones harder to anticipate. My steering experiments also showed how fragile representation-level interventions can be. I want to evaluate interventions not only by how often they succeed, but by whether their remaining failures stay detectable, and to use interpretability to build monitors that can still catch them.

More broadly, I want to help make model reliability something we can verify, rather than something we hope prompting or training has achieved.

References

1. W. H. Holliday, M. Mandelkern, C. E. Zhang. Conditional and modal reasoning in large language models. EMNLP, 2024.
2. **F. Shahrokhshahi**, F. Mohammadi, F. Sonmez. LogiCue: Targeted prompting for improved modal and conditional reasoning in large language models. Procedia Computer Science 275, pp. 484–492, 2026. Paper · Code
3. **F. Shahrokhshahi**, F. Mohammadi. Can we predict LLM reasoning failures? Structural predictability of modal and conditional inference errors. In: KI 2026: Advances in Artificial Intelligence, LNAI vol. 16830, pp. 317–323. Springer, 2026. Paper · Code
4. **F. Shahrokhshahi**. Towards logical reasoning by design: Encoding inference rules in MeMo's associative memory. MeMo Workshop on Mechanistic Interpretability & Neuro-symbolic Approaches by-design, Rome, 2026. CEUR Workshop Proceedings, to appear. Code
5. F. M. Zanzotto et al. MeMo: Towards language models with associative memory mechanisms. Findings of ACL, 2025.
6. A. M. Turner et al. Activation addition: Steering language models without optimization. arXiv:2308.10248, 2023.
7. D. Gurgurov et al. CLaS-Bench: A cross-lingual alignment and steering benchmark. Findings of ACL, 2026.

Other code: cross-domain steering · RAG system